MANUFACTURING DOWNTIME ANALYSIS & PREDICTION

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Business Data Analysis Report

I. Project Overview

TThe project aims to analyze manufacturing line productivity and downtime patterns to help decision-makers improve operational efficiency and production planning.

Using the dataset provided, the project focuses on cleaning and preprocessing production data, identifying key factors affecting downtime, generating analytical insights, and building a predictive model to forecast the next day's downtime. Based on the predicted downtime, the project will estimate the optimal number of production batches.

Project Objectives:

- Clean and preprocess manufacturing data using **Power BI**
- Identify key drivers of downtime
- Generate business-oriented analytical questions
- Predict next-day downtime
- Recommend batch production quantities based on predicted downtime
- Build an interactive dashboard for decision-makers
- Present final insights in a professional report and presentation

II. Project Methodology:

A. Week 1 – Data Cleaning & Preprocessing

1. Data validation and handling missing values
2. Removing inconsistencies and duplicates
3. Data transformation and feature preparation
4. Deliverables:
    - Clean dataset
    - Preprocessing documentation

B. Week 2 – Analysis PhaseDevelop business-driven analysis questions

- Explore downtime patterns (shift, operator, machine, date, etc.)
- Build predictive logic for next-day downtime
- Estimate number of production batches accordingly
- Deliverables:
    - Set of analytical questions
    - Downtime prediction logic

Week 3 – Dashboard Development

- Build interactive Power BI dashboard
- Visualize KPIs:
    - Total Downtime
    - Downtime by Machine
    - Downtime by Shift
    - Downtime Trend
    - Predicted Downtime
    - Recommended Batch Count
- Deliverable:
    - Fully functional Power BI Dashboard

Week 4 – Final Presentation

- Final report summarizing:
    - Data preparation
    - Key insights
    - Prediction model
    - Business recommendations
- Professional presentation slides

GitHub Link